Automating Data Import and Relationship Mapping Using Import Sets in ServiceNow

1. Introduction

Modern organizations manage large volumes of data from multiple sources such as spreadsheets, databases, and external systems. Manually importing and organizing this data can be time-consuming and error-prone. ServiceNow provides a powerful mechanism called **Import Sets** to automate the process of importing external data into ServiceNow tables.

This project focuses on automating the **data import process and relationship mapping** using ServiceNow Import Sets. By leveraging transform maps, staging tables, and automated scripts, the system ensures that imported data is correctly structured and related to existing records in the ServiceNow database.

The automation improves efficiency, reduces manual work, and ensures data consistency across the platform.

2. Objectives

The main objectives of this project are:

- To automate the process of importing external data into ServiceNow.

- To utilize **Import Sets and Transform Maps** for structured data transformation.
- To establish relationships between imported records and existing ServiceNow tables.
- To reduce manual intervention in data management.
- To improve data accuracy and consistency.

3. Overview of ServiceNow Import Sets

An **Import Set** in ServiceNow acts as a staging area where external data is first imported before being transformed into target tables.

The process typically includes the following steps:

1. Data is imported into a **staging table**.
2. A **Transform Map** is used to define how data should be transferred.
3. Data is moved into the **target table**.
4. Relationships between records are automatically created.

Import Sets are commonly used when importing data from:

- CSV files
- Excel spreadsheets
- External databases
- Third-party applications

This mechanism ensures that raw data is processed and validated before being inserted into production tables.

4. System Architecture

The architecture of this project consists of the following components:

1. Data Source

External data file (CSV/Excel) containing information that needs to be imported.

2. Import Set Table

Temporary staging table where imported data is stored before transformation.

3. Transform Map

Defines how data fields from the Import Set table map to the target table.

4. Target Table

Final ServiceNow table where processed records are stored.

5. Relationship Mapping

Ensures the imported data is linked correctly with existing records.

Workflow

External File → Data Source → Import Set Table → Transform Map → Target Table → Relationship Mapping

5. Tools and Technologies Used

Tool	Purpose
ServiceNow Platform	Main development environment
Import Sets	Staging table for external data
Transform Maps	Field mapping and data transformation
CSV/Excel Files	Source data format
Scripts	Automating transformation logic
GitHub	Version control and documentation

6. Implementation Steps

Step 1: Create Data Source

A data source was created in ServiceNow to upload the external data file.

Steps:

1. Navigate to **System Import Sets → Data Sources**
2. Create a new data source
3. Upload CSV/Excel file
4. Specify file format and delimiter

The system automatically creates an **Import Set Table** for the uploaded data.

Step 2: Load Data into Import Set

After creating the data source, the data was loaded into the staging table.

Steps:

1. Click **Load All Records**
2. ServiceNow imports the file data
3. Data is stored in the Import Set table

At this stage, data is not yet inserted into the final table.

Step 3: Create Transform Map

A **Transform Map** was created to define how data fields should be transferred.

Steps:

1. Navigate to **System Import Sets → Transform Maps**
2. Create a new transform map
3. Select:
    - Source Table (Import Set Table)
    - Target Table

Example Mapping:

Source Field	Target Field
User_Name	name
Email	email
Department	department
Manager	Manager

Step 4: Field Mapping

Field mapping connects columns from the import set to columns in the target table.

For example:

CSV Field → ServiceNow Field

Name → sys_user.name
Email → sys_user.email
Department → cmn_department.name

This ensures proper data transfer.

Step 5: Relationship Mapping

Relationship mapping ensures that imported records connect with existing records.

Examples:

- User → Department
- User → Manager
- Asset → Configuration Item

This was achieved using **Reference Fields and Transform Scripts**.

Example script logic:

- Check if department exists
- If exists → link record
- If not → create new department

This maintains correct relationships within ServiceNow tables.

7. Automation Using Transform Scripts

Transform scripts were used to automate data validation and relationship mapping.

Example functions implemented:

OnBefore Script

Validates data before insertion.

Example tasks:

- Check duplicate records
- Validate email format

OnAfter Script

Performs actions after transformation.

Example tasks:

- Assign default roles
- Update related records

These scripts help automate the entire import process.

8. Testing and Validation

Testing was performed to ensure the import process worked correctly.

Test cases included:

Test Case	**Expected Result**
Upload valid CSV file	Data imported successfully
Missing fields	Validation error shown
Duplicate records	Duplicate prevented
Relationship mapping	Correct references created

Results showed that the system successfully automated data import and maintained relationships between records.

9. Benefits of the Solution

This automated solution provides several benefits:

1. Time Efficiency

Reduces manual effort required for data entry.

2. Data Accuracy

Minimizes errors during data import.

3. Scalability

Can handle large datasets easily.

4. Automation

Reduces dependency on manual processes.

5. Data Consistency

Maintains correct relationships between records.